

Fuzzy Quantum Group Decision Making and Its Application in Meteorological Disaster Emergency

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Abstract—This article proposes a quantum group decision model that integrates intuitionistic fuzzy sets to represent the uncertainty of meteorological disaster information, addressing both vagueness and probabilistic uncertainty. This makes it particularly suitable for modeling the complex and dynamic decision-making processes during emergency responses. The model employs regret theory for attribute weight determination and constructs a quantum-like Bayesian network (QLBN), where Deng entropy is applied to measure the mutual interference effects among decision-makers. Decision-makers' weights are determined using grey relational analysis and incorporated as the initial layer in the Bayesian network. The conditional probabilities within the QLBN are derived by integrating attribute weights and regret utility functions, and the alternatives are ranked based on their final quantum probabilities. The effectiveness and stability of the model are demonstrated through its application in emergency alternative selection for meteorological disasters, confirmed by sensitivity and comparison analyzes.

Index Terms—Deng entropy, grey relational analysis, quantum group decision model, quantum-like Bayesian network (QLBN), regret theory.

I. INTRODUCTION

MULTI-ATTRIBUTE group decision making (MAGDM) constitutes a pivotal research area within decision science, playing a significant role in various practical applications [1], [2], [3]. Particularly in the context of meteorological disaster emergency rescue, multiple departments need to be participated, and multiple rescue indicators should be considered. This collaboration exemplifies the typical MAGDM application scenarios [4], [5]. For instance, during natural disasters such as

floods or typhoons, proper MAGDM mechanism is crucial for ensuring the rapidity and efficiency of the response efforts [6].

In the context of meteorological disasters, decision-makers are often required to make prompt judgments in uncertain environment, thereby amplifying the challenges in gathering and coping with information efficiently. There are great difficulties in obtaining accurate and timely data. The information is represented by various forms, including crisp numbers [7], [8], fuzzy set [9], [10], linguistic term set [11], [12], and grey number [13], and so on. Intuitionistic fuzzy set (IFS) can fully express the uncertainty of information through the dual description of membership degree and nonmembership degree, and hesitation degree is introduced to quantify the information shortage or conflict in the decision-making process. This feature is especially suitable for complex and dynamic uncertain information processing in disaster emergency decision-making scenarios. In addition, compared with type 1 fuzzy set, type 2 fuzzy set, and hesitant fuzzy set, IFS is more suitable for uncertain fuzzy decision environment, because it can express the membership degree and nonmembership degree of decision maker at the same time, and fully reflect the fuzzy judgment of decision maker. Type 1 fuzzy sets can only represent a single membership degree and lacks a comprehensive description of uncertainty. Although Type 2 fuzzy sets have better expressiveness in dealing with uncertainty and fuzziness, their computational complexity is higher and their model construction is more complex. Hesitant fuzzy sets focus on the decision maker's hesitation, but its expression ability is limited. Therefore, IFSs are particularly effective as a method for representing information to assist decision-makers in contexts of uncertain and fuzzy decision-making, such as disaster emergency response scenarios [14], [15], [16].

In MAGDM problems, the attribute weighting is crucial for ensuring the accuracy and reliability of decision results. Although methods such as the AHP method, entropy weight method, and TOPSIS method have been extensively applied to address the issues [17], [18], [19], [20], the influence of decision makers' psychological factors and behavioral preferences on weight determination and decision rationality should not be ignored. Regret behavior is a common phenomenon in decision-making, and the regret theory reflecting this behavior has been widely used [21], [22], [23]. For instance, Liang and Wang [24] introduced a probabilistic interval-valued hesitation fuzzy gain or loss advantage scoring method that leverages regret theory and interval evidence reasoning to navigate uncertain information in complex decision environments. Similarly, Zhang et al. [25] devised a multigranularity three-state decision model rooted in

Received 10 October 2024; revised 10 December 2024; accepted 29 December 2024. Date of publication 2 January 2025; date of current version 5 May 2025. This work was supported in part by the Social Science Fund Project of Jiangsu Province under Grant 23GLB006, in part by the National Natural Science Foundation of China under Grant 72371137 and Grant 71801085, in part by the Jiangsu University Philosophy and Social Science project under Grant 2022SJYB0187, in part by the National Scholarship Foundation of China under Grant 202209045002, in part by MCIN/AEI/10.13039/501100011033 under PID2022-139297OBI00, in part by "ERDF A way of making Europe", in part by the Regional Ministry of University, Research and Innovation, and in part by the European Union under the Andalusia ERDF Programme under Grant 2021-2027 C-ING-165-UGR23. Recommended by Associate Editor A. S. Albahri. (Corresponding author: Zaiwu Gong.)

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Digital Object Identifier 10.1109/TFUZZ.2024.3525009

regret theory to mirror decision-makers' bounded rationality, and Zhu et al. [26] developed a three-way consistency model employing regret theory, where a minimal adjustment feedback mechanism aims to balance consistency with individual independence. Overall, incorporating regret theory into the weight determination process allows for a more comprehensive consideration of decision-makers' psychological factors and behavioral preferences, leading to weight allocations that better reflect actual decision-making scenarios.

The opinions of decision-makers in a group decision-making environment are usually interdependent. However, existing studies on the aggregation of group opinions often do not consider the mutual interference effect [27]. Consequently, traditional decision models based on classical probability struggle to effectively represent this situation. Quantum probability theory (QPT) enhances classical probability theory by offering a more intricate representation of the interference among entities in intricate systems. This method can reflect the paradoxes, sub-optimal choices and irrational behaviors often observed in the decision-making process [28], [29], the interference effects affect probability calculations. Therefore, decision analysis based on quantum theoretical framework presents considerable advantages [30], [31]. For instance, Ji et al. [32] employed overlapping nodes to simplify trust propagation and incorporated the concept of trust attenuation to achieve efficient consensus. However, they did not consider the mutual interference effect between nodes. In contrast, the quantum-guided expert state transformation, alongside the minimum cost consistency model and attributes' weights optimization model developed through the normal cloud model, adeptly navigates changes and the complexities of achieving group consensus [33]. Wang et al. [34] examined how social networks affect the development of group decision views and introduced a trust recognition model that relies on quantum interference. These models collectively address the issue of subjective beliefs concerning the independence or interrelatedness of decision-makers in MAGDM. By conceptualizing decision-makers' viewpoints as simultaneous wave functions, this method simulates the interference of beliefs, thereby impacting the ultimate decision results [35], [36], [37], utilizing quantum-like Bayesian network (QLBN) and belief entropy to align with human cognitive processes regarding the interference relationships between attributes, belief entropy serves as a metric for quantifying probabilistic uncertainty, facilitating the computation of disturbance values within the quantum framework [38]. Thus, this process improves the quality of decision-making and guarantees that outcomes are thorough and precise [39], [40].

In the MAGDM framework, recognizing and measuring the interactions among various decision alternatives is very important. By measuring the relationship between different sequences, grey relational analysis reflects the characteristics of proximity and similarity between sequences, and is widely used in alternative selections [41]. Grey relational analysis is to draw the discrete data sequence representing the behavior of the system factors as a continuous line graph, and reflect the proximity according to the geometric features of the sequence line graph, and judge the ranking of the sequences according to the proximities between the element sequences and the feature sequence [42].

The main motivations of this article are as follows.

Motivation 1: Conventional MAGDM models often assume that decision makers' opinions are independent of each other, which is not always realistic. In practice, decision makers' beliefs frequently influence one another, leading to interference effects that are difficult to manage. In addition, decision-making in real-world scenarios, especially in uncertain environments, such as meteorological disasters, often involves vague and imprecise information. The fuzzy quantum model addresses these challenges by integrating fuzzy sets to capture the vagueness in human judgment and quantum decision theory to effectively handle mutual interference between decision makers. This combination provides a more accurate and flexible framework for decision-making in complex, uncertain environments.

Motivation 2: The measurement of the interference value of the decision maker is a difficult problem, especially in determining the phase angle difference between two decision makers. In most studies, the range of phase angle is set within $[0, 2\pi]$ to analyze the influence of the change of interference value on the probability of alternatives, but this method lacks accuracy [30], [31]. Therefore, the concept of Deng entropy is introduced to simulate the phase angle change, so as to improve the prediction ability and practicability of the model.

This article's primary contributions are outlined as follows.

- 1) A multiattribute fuzzy quantum group decision model (MAFQGDM) is constructed that not only captures the mutual interference effects between decision makers' perspectives but also incorporates fuzzy sets to address the vagueness and imprecision inherent in real-world decision-making processes. This method allows the model to closely mirror the complexity and uncertainty of decision-making in practical applications, offering a more robust and flexible framework.
- 2) A new scoring function is proposed that translates intuitionistic fuzzy numbers into real values, factoring in membership, nonmembership, and hesitation elements to improve the model's intuitiveness and computational efficiency.
- 3) Using grey relational analysis to explore the relationship between individual decision maker's evaluations and the overall assessment, to determine decision makers' weights and obtain key nodes' probabilities in the quantum-like Bayesian framework.
- 4) Adopting the regret theory to get attributes' weights reflecting psychological factors, and obtain conditional probabilities in QLBNs.

In short, in uncertain and complex decision-making problems, such as emergency response to meteorological disasters, the decision information is difficult to quantify and highly uncertain, and the interference effect among decision makers is usually accompanied by high uncertainty and information fuzziness. The quantum decision framework can effectively deal with the interference effect among decision makers. Therefore, this article introduces IFS as an information representation method and combines it with quantum decision framework to better adapt to complex fuzzy decision scenes. In this article, a new intuitionistic fuzzy score function is proposed, which considers

membership, nonmembership and hesitation, and introduces risk coefficient to better deal with the incompleteness of decision information and the uncertainty of decision maker's risk attitudes. At the same time, according to the principle of grey correlation analysis to measure the proximity and similarity of two sequences in uncertain environment, the weights of decision makers are determined. Considering the regret psychology of decision makers, according to the regret utility value of each scheme under different attributes, an optimization model is constructed to determine the attributes' weights. These innovations fully reflect the decision-making process under uncertain environment.

The rest of this article is organized as follows. Section II reviews the framework of IFS, regret theory, Deng entropy, and QPT. In Section III, an MAFQGDM is proposed. The model is constructed by combining regret theory, Deng entropy, quantum probability, and Bayesian networks. Section IV demonstrates the use of MAFQGDM in emergency rescue for meteorological catastrophes. The stability and reliability of the model are validated using sensitivity analysis and comparison analysis. Section V ultimately presents a concise overview of the primary discoveries and contributions, while also proposing potential avenues for further study. Finally, Section V concludes this article.

II. PRELIMINARIES

The purpose of this part is to examine and summarize the fundamental theories, establishing a strong theoretical basis for further discussions in the following sections. It includes IFS, Deng entropy, regret theory, and QPT in decision making. Through in-depth discussion of these theories, a comprehensive theoretical framework is constructed to provide necessary theoretical support for the analysis and research of the following chapters.

A. Intuitionistic Fuzzy Set

Define the domain X as a finite nonempty set, and $Q = \{(x, \mu_Q(x), v_Q(x)) | x \in X\}$ as an IFS within X , where $\mu_Q(x), v_Q(x) : x \rightarrow [0, 1]$ separately denotes the membership and nonmembership functions of Q . For any element in $x \in X$, it satisfies condition $0 \leq \mu_Q(x) + v_Q(x) \leq 1$. Furthermore, the hesitancy degree for an element x belonging to set Q is $\pi_Q(x) = 1 - \mu_Q(x) - v_Q(x)$.

Definition 1 (see [44]): Let $\alpha_1 = (\mu_{\alpha_1}, v_{\alpha_1})$ and $\alpha_2 = (\mu_{\alpha_2}, v_{\alpha_2})$ be two intuitionistic fuzzy numbers, then $\alpha_1 = \alpha_2$, if and only if $\mu_{\alpha_1} = \mu_{\alpha_2}, v_{\alpha_1} = v_{\alpha_2}$.

For simplicity, let $\alpha = (\mu_\alpha, v_\alpha)$ be the intuitionistic fuzzy number, $\mu_\alpha \in [0, 1]$ be the membership degree, $v_\alpha \in [0, 1]$ be the nonmembership degree, and $\mu_\alpha + v_\alpha \leq 1$. The hesitancy degree of the intuitive fuzzy number α is denoted as $\pi_\alpha = 1 - \mu_\alpha - v_\alpha$.

In IFS theory, the score and exact functions are essential for comparing and ranking intuitionistic fuzzy numbers. The score function evaluates the overall information of these numbers by comparing their values. A higher score indicates greater membership and lower nonmembership. Conversely, the exact

function measures uncertainty, a lower value signifies less uncertainty and more accurate information [45], [46].

These functions quantify the membership degree, nonmembership degree, and uncertainty of intuitionistic fuzzy numbers, aiding in effective evaluation and ranking in MAGDM and fuzzy decision support systems. The score function assesses the total information, while the exact function evaluates information uncertainty.

Definition 2 (see [47]): Let $\alpha = (\mu_\alpha, v_\alpha)$ be an intuitionistic fuzzy number, then

$$s(\alpha) = \mu_\alpha - v_\alpha. \quad (1)$$

$$h(\alpha) = \mu_\alpha + v_\alpha. \quad (2)$$

where $s(\alpha)$ and $h(\alpha)$ are the scoring function and the exact function of α , respectively, and $s(\alpha) \in [-1, 1], h(\alpha) \in [-1, 1]$.

For any two intuitionistic fuzzy numbers α and β , then

- 1) if $s(\alpha) < s(\beta)$, then $\alpha \prec \beta$;
- 2) if $s(\alpha) = s(\beta)$, then $h(\alpha) > h(\beta), \alpha \succ \beta$; if $h(\alpha) = h(\beta), \alpha = \beta$, then $h(\alpha) < h(\beta), \alpha \prec \beta$.

B. Deng Entropy

Deng entropy, often referred to as belief entropy, is a concept that elucidates the mechanism via which a system undergoes a transformation from one state to another [48].

Definition 3 (see [49]): For given mass function m defined on frame of discernment $\Theta = \{\theta_1, \theta_2, \dots, \theta_N\}$, Deng entropy is defined as follows:

$$E_D(m) = - \sum_{Y \in 2^\Theta} m(Y) \log_2 \left[\frac{m(Y)}{2^{|Y|} - 1} \right]. \quad (3)$$

where Y is the focus element.

C. Regret Theory

Regret theory [50], [51] is a decision theory that studies how people consider potential regret when making choices in the face of uncertainty. The theory assumes that decision makers not only focus on possible outcomes, but also weigh the degree of regret that might result if they choose different options, thus influencing their final decision.

Based on the regret theory [50], [51], let $v(\hat{x})$ and $v(\hat{y})$ represent the utility that the decision maker may get from the outcomes of option A and B , respectively. Let $R(v(\hat{x}) - v(\hat{y}))$ denote the regret-joy value. In this context, the perceived utility of decision makers for choice A can be expressed as

$$v(\hat{x}, \hat{y}) = v(\hat{x}) + R(v(\hat{x}) - v(\hat{y})). \quad (4)$$

And if $R(v(\hat{x}) - v(\hat{y})) > 0$, the decision maker is happy to choose plan A . When $R(v(\hat{x}) - v(\hat{y})) < 0$, decision makers choose B . Here, the regret-joy function $R(v(\hat{x}) - v(\hat{y}))$ is a monotonically increasing concave function [50], [51], [52], satisfying $R'(v(\hat{x}) - v(\hat{y})) > 0$ and $R''(v(\hat{x}) - v(\hat{y})) < 0$, and $R(0) = 0$.

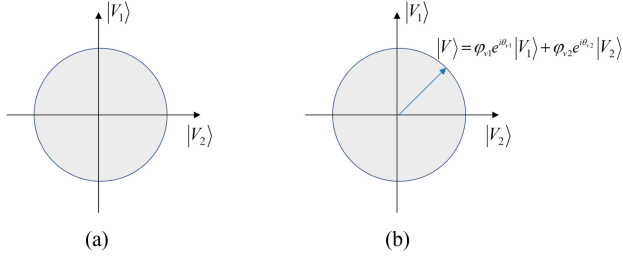


Fig. 1. Events in Hilbert space. (a) Hilbert space in QPT. (b) Event in the Hilbert space.

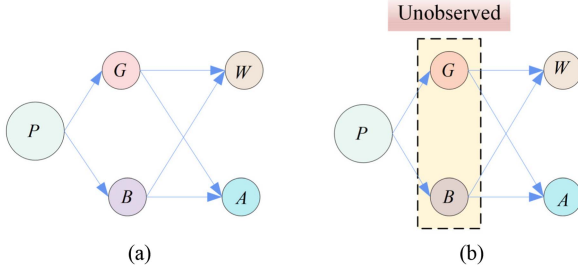


Fig. 2. Markov chain path diagram. (a) Markov process. (b) Quantum process.

D. Quantum Probability Theory

In QPT, compared to classical probability theory, all potential occurrences may be regarded as subspaces of Hilbert spaces, as shown in Fig. 1.

The vector space V is formed by a collection of orthogonal basis vectors, denoted as $V = \{|V_1\rangle, \dots, |V_n\rangle\}$. In Hilbert space, events are defined using superposition of quantum states $|D\rangle$, which are expressed as linear combinations of orthogonal basis vectors

$$\begin{aligned} |D\rangle &= \langle V_1 | D \rangle |V_1\rangle + \langle V_2 | D \rangle |V_2\rangle + \dots + \langle V_n | D \rangle |V_n\rangle \\ &= \psi_1 |V_1\rangle + \psi_2 |V_2\rangle + \dots + \psi_n |V_n\rangle \\ &= \varphi_1 e^{i\theta_1} |V_1\rangle + \varphi_2 e^{i\theta_2} |V_2\rangle + \dots + \varphi_n e^{i\theta_n} |V_n\rangle. \end{aligned} \quad (5)$$

$|\varphi_k|^2$ is directly correlated with the observed probability of that state, following Born's rule:

$$\begin{aligned} P(V_k) &= |\varphi_k e^{i\theta_k}|^2 \\ &= \varphi_k e^{i\theta_k} \times \varphi_k e^{-i\theta_k} = (\varphi_k)^2, (k = 1, 2, \dots, n). \end{aligned} \quad (6)$$

E. Decision Making in QPT

QPT, rooted in quantum mechanics, employs wave functions to represent the state of a system and describes probabilities as the square of their amplitudes. Key concepts, such as superposition and entanglement, highlight the intrinsic uncertainty of states and the interdependence of measurements, providing explanations for certain nonclassical behaviors in human decision-making, as shown in Fig. 2 [28].

Fig. 2(a) shows the expression of a classical probability process, describing the total probability of event A occurring

$$P(A) = P(G)P(A|G) + P(B)P(A|B). \quad (7)$$

Quantum probabilistic processes are depicted in Fig. 2(b). The probability may be described by its probability amplitude, and its magnitude is equal to the projection length squared. Feynman's second rule states that you must square the total of the probability amplitudes on all potential pathways in order to get the probability of the final transition if you are unable to see the precise path of the state change. The final quantum probability of event A is as follows [53]:

$$\begin{aligned} P(A) &= P(G)P(A|G) + P(B)P(A|B) \\ &\quad + 2\sqrt{P(G)P(A|G)P(B)P(A|B)} \cos(\alpha - \beta). \end{aligned} \quad (8)$$

F. Novel IFS Scoring Function

The scoring function described in (1) under Definition 4 incorporates the membership, nonmembership degrees but overlooks the hesitation degree's impact on decision outcomes. This omission could lead to discrepancies between the scoring function's application in ranking schemes and actual scenarios. Drawing on previous research and considering this limitation, an enhanced scoring function is hereby defined [54], [55], [56].

Definition 4: Let $\alpha = (\mu_\alpha, v_\alpha)$ be an intuitionistic fuzzy number, then the scoring function is defined

$$s(\alpha) = u_\alpha - v_\alpha - \kappa \ln(1 - \pi_\alpha). \quad (9)$$

where $\pi_\alpha = 1 - \mu_\alpha - v_\alpha$, then the scoring function can be expressed as

$$s(\alpha) = u_\alpha - v_\alpha - \kappa \ln(2 - u_\alpha - v_\alpha). \quad (10)$$

The score function $s(\alpha) \in [-1, 1]$, risk coefficient $\kappa \in [0, 1]$, and hesitancy reflect the incompleteness of decision information or the uncertainty of decision makers. Therefore, the risk coefficient can help the score function establish a reasonable balance between these different dimensions of information and ensure that the score can comprehensively reflect all aspects of the intuitionistic fuzzy information. And when $\kappa = 0$, (9) falls back to (1). But the psychology of decision makers is essentially risk-averse, so this is rare. In summary, this scenario is an edge case rather than the norm. The flexibility provided by κ ensures that the score function can be tailored to different risk sensitivities, preserving its discriminative power and enhancing its practical applicability.

Theorem 1: Let $\alpha = (\mu_\alpha, v_\alpha)$ be intuitionistic fuzzy number, and $s(\alpha)$ be the scoring function of α , then:

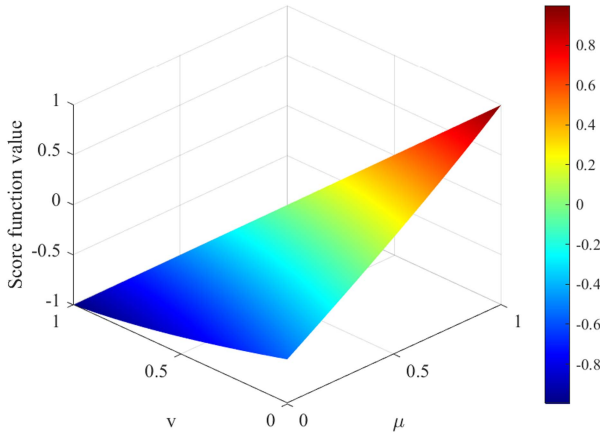
- 1) $\mu_\alpha = 1, v_\alpha = 0 \Leftrightarrow s(\alpha) = 1$;
- 2) $\mu_\alpha = 0, v_\alpha = 1 \Leftrightarrow s(\alpha) = -1$.

Proof:

- 1) When $\mu_\alpha = 1, v_\alpha = 0$, according to the definition of score function can get $s(\alpha) = 1$.
- 2) When $\mu_\alpha = 0, v_\alpha = 1$, according to the definition of score function can get $s(\alpha) = -1$. $\square$

Theorem 2: Let $\alpha = (\mu_\alpha, v_\alpha)$ be intuitionistic fuzzy number, then the score function $s(\alpha)$ of α increases monotonically with respect to μ_α and decreases monotonically with respect to v_α .

Proof: First, $\frac{\partial s(\alpha)}{\partial \mu_\alpha} = 1 + \frac{\kappa}{2 - \mu_\alpha - v_\alpha} \geq 0$. Second, $\frac{\partial s(\alpha)}{\partial v_\alpha} = -1 + \frac{\kappa}{2 - \mu_\alpha - v_\alpha} \leq 0$. That is, the scoring function satisfies the monotonicity theorem. $\square$


 Fig. 3. Variation distribution of novel score function $s(\alpha)$.

Theorem 3: Let $\alpha_1 = (\mu_{\alpha_1}, v_{\alpha_1})$ and $\alpha_2 = (\mu_{\alpha_2}, v_{\alpha_2})$ be two intuitionistic fuzzy numbers, if $\mu_{\alpha_1} > \mu_{\alpha_2}$ and $v_{\alpha_1} < v_{\alpha_2}$, then $s(\alpha_1) > s(\alpha_2)$.

Proof:

$$\begin{aligned} s(\alpha_1) - s(\alpha_2) &= [\mu_{\alpha_1} - v_{\alpha_1} - \kappa \ln(1 - \pi_{\alpha_1})] - [\mu_{\alpha_2} - v_{\alpha_2} - \kappa \ln(1 - \pi_{\alpha_2})] \\ &= [(\mu_{\alpha_1} - v_{\alpha_1}) - (\mu_{\alpha_2} - v_{\alpha_2})] + \kappa [\ln(1 - \pi_{\alpha_2}) - \ln(1 - \pi_{\alpha_1})] \\ &= [(\mu_{\alpha_1} - v_{\alpha_1}) - (\mu_{\alpha_2} - v_{\alpha_2})] + \kappa \ln \frac{2 - \mu_{\alpha_2} - v_{\alpha_2}}{2 - \mu_{\alpha_1} - v_{\alpha_1}} \end{aligned}$$

where $(\mu_{\alpha_1} - v_{\alpha_1}) > (\mu_{\alpha_2} - v_{\alpha_2})$ and $2 - \mu_{\alpha_2} - v_{\alpha_2} > 2 - \mu_{\alpha_1} - v_{\alpha_1}$, so $\ln \frac{2 - \mu_{\alpha_2} - v_{\alpha_2}}{2 - \mu_{\alpha_1} - v_{\alpha_1}} > 0$, that is $s(\alpha_1) > s(\alpha_2)$. $\square$

Fig. 3 shows the variation of the score function of intuitionistic fuzzy number under different membership degrees and non-membership degrees.

In order to verify the validity of the score function in this article, a comparative analysis of the score function is carried out. First, the classical score function [47] is not flexible enough in the face of data with different hesitancy and fuzziness, that is, when the membership degree and nonmembership degree are the same, the classical score function cannot distinguish effectively, because it fails to take into account the influence of hesitancy degree. Second, compared with an extended score function [55], although it takes into account the influence of the degree of hesitation, it lacks a flexible adjustment mechanism for the degree of hesitation. On this basis, this article compares and analyzes the score function based on the score values of 5 different groups. The specific data is as follows.

- 1) *Group 1:* $\mu_{\alpha} = 0.5, v_{\alpha} = 0.5, \pi_{\alpha} = 0$.
- 2) *Group 2:* $\mu_{\alpha} = 0.4, v_{\alpha} = 0.4, \pi_{\alpha} = 0.2$.
- 3) *Group 3:* $\mu_{\alpha} = 0.3, v_{\alpha} = 0.2, \pi_{\alpha} = 0.5$.
- 4) *Group 4:* $\mu_{\alpha} = 0.2, v_{\alpha} = 0.4, \pi_{\alpha} = 0.4$.
- 5) *Group 5:* $\mu_{\alpha} = 0.1, v_{\alpha} = 0.1, \pi_{\alpha} = 0.8$.

The comparison results are shown in Fig. 4.

By analyzing the score ordering changes of these arrays, it can be found that with the change of parameter κ , the score ordering

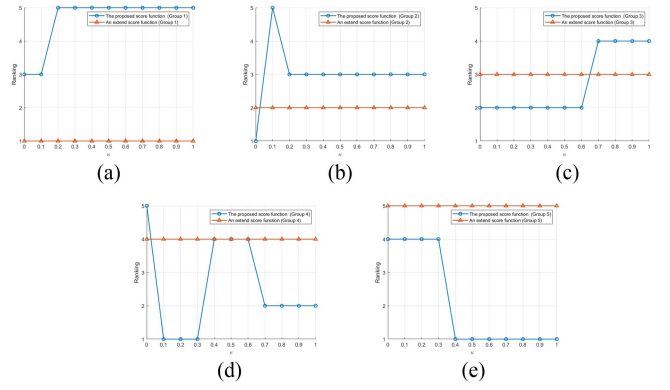


Fig. 4. Comparative analysis of sorting results of different scoring functions in different arrays. (a) Group 1. (b) Group 2. (c) Group 3. (d) Group 4. (e) Group 5.

results of each array will change significantly. Specifically, the different values of κ have a significant impact on the sorting of arrays, especially in the face of high hesitancy, the scoring function can flexibly adjust the sorting result of the scheme. Therefore, the score function has stronger adaptability, and can dynamically adjust the decision result according to the different decision background, so as to better meet the needs of practical applications.

III. MULTIATTRIBUTE FUZZY QUANTUM GROUP DECISION MAKING MODEL CONSIDERING INTERFERENCE EFFECT

In this section, this article will quantify the interference effect in the MAGDM process with IFS, based on this, a novel fuzzy quantum framework is proposed.

The MAGDM problem consists of decision group $D' = \{d_1, d_2, \dots, d_s\}$, alternative set $A' = \{a_1, a_2, \dots, a_m\}$, attribute set $C = \{c_1, c_2, \dots, c_n\}$. Where attributes' weights set is $W = \{\omega_1, \omega_2, \dots, \omega_n\}$, decision makers' weights set is $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$, r_{ij}^k is the evaluation value of decision maker d_k ($k = 1, \dots, s$) about attribute c_j ($j = 1, \dots, n$) of alternative a_i ($i = 1, \dots, m$).

A. Collecting Decision Information

1) Evaluation information collection: $r_{ij}^k = (u_{\alpha}^k, v_{\alpha}^k)$ is represented by the Intuitionistic fuzzy number, and the evaluation information matrix is represented by $R^{(k)} = (r_{ij}^k)_{m \times n}$

$$R^{(k)} = \begin{pmatrix} r_{11}^k & r_{12}^k & \cdots & r_{1n}^k \\ r_{21}^k & r_{22}^k & \cdots & r_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1}^k & r_{m2}^k & \cdots & r_{mn}^k \end{pmatrix}, k = 1, \dots, s. \quad (11)$$

Based on the new score function (9), the initial information evaluation matrix is transformed into score function matrix

$$\tilde{R}^{(k)} = (s(r_{ij}^k))_{m \times n}$$

$$\tilde{R}^{(k)} = \begin{pmatrix} s(r_{11}^k) & s(r_{12}^k) & \cdots & s(r_{1n}^k) \\ s(r_{21}^k) & s(r_{22}^k) & \cdots & s(r_{2n}^k) \\ \vdots & \vdots & \ddots & \vdots \\ s(r_{m1}^k) & s(r_{m2}^k) & \cdots & s(r_{mn}^k) \end{pmatrix}, k = 1, \dots, s. \quad (12)$$

2) The data must be normalized in order to remove the influence of various qualities, such as cost or benefit types, which might create incommensurability. The normalized form is represented as $\hat{r}_{ij}^k$

$$\hat{r}_{ij}^k = \begin{cases} \frac{s(r_{ij}^k) - \min_i s(r_{ij}^k)}{\max_i s(r_{ij}^k) - \min_i s(r_{ij}^k)}, c_j \text{ for benefit} \\ \frac{\max_i s(r_{ij}^k) - s(r_{ij}^k)}{\max_i s(r_{ij}^k) - \min_i s(r_{ij}^k)}, c_j \text{ for cost} \end{cases} \quad (13)$$

$i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s.$

B. Decision Makers' Weighting Based on Grey Relational Analysis

The relationships among alternatives in the MAFQGDM problem are often interrelated and mutually restrictive. Grey relational analysis effectively quantifies this alignment by comparing each decision-maker's evaluations against the optimal reference. Decision-makers whose evaluations closely match the optimal alternative are assigned higher weights, indicating that their assessments are more accurate and comprehensive. Based on this premise, this article utilizes the grey relational analysis method to establish the grey relational degree between the alternatives evaluated by each decision-maker and the optimal alternative.

The primary objective of grey relational analysis is to determine the similarity between the initial data series by calculating the grey relational coefficient and grey relational degree. Therefore, using initial data rather than standardized data aligns with the core principle of grey relational analysis. This approach simplifies the process and preserves the original data schema, ensuring an accurate measure of the relationship degree without unnecessary transformations.

The optimal score for each attribute under each decision-maker is used as the reference sequence $S_0^k(j)$. The correlation between each decision-maker's behavioral sequence $S_i^k(j)$ and this reference sequence is calculated individually [41]. The weights of the decision-makers are determined based on grey relational degree and are then translated into probabilities for the first layer of the QLBN.

First, the reference sequence is defined as

$$S_0^k(j) = [s_0^k(1), s_0^k(2), \dots, s_0^k(n)], k = 1, \dots, s. \quad (14)$$

where

$$s_0^k(j) = \max_i s(r_{ij}^k), k = 1, \dots, s, j = 1, \dots, n. \quad (15)$$

Calculate the difference between the behavioral sequence $S_i^k(j)$ and the reference sequence $S_0^k(j)$ for each decision maker

$$\Delta_i^k(j) = |s_0^k(j) - s(r_{ij}^k)|, i = 1, \dots, m, j = 1, \dots, n, k = 1, \dots, s. \quad (16)$$

The difference sequence can be expressed as

$$\Delta_i^k = [\Delta_i^k(1), \Delta_i^k(2), \dots, \Delta_i^k(n)], i = 1, \dots, m, k = 1, \dots, s. \quad (17)$$

Calculate the grey relational coefficient

$$\gamma_{0i}^k(j) = \frac{\min_j \min_i \Delta_i^k(j) + \xi \max_j \max_i \Delta_i^k(j)}{\Delta_i^k(j) + \xi \max_j \max_i \Delta_i^k(j)} \quad (18)$$

$i = 1, \dots, m, j = 1, \dots, n, k = 1, \dots, s$

where ξ is the resolution coefficient, $\xi \in [0, 1]$.

Calculate grey relational degree

$$\gamma_{0i}^k = \frac{1}{n} \sum_{j=1}^n \gamma_{0i}^k(j), i = 1, \dots, m, k = 1, \dots, s. \quad (19)$$

On this basis, the weights of decision makers are calculated as follows:

$$\lambda_k = \frac{\sum_{i=1}^m \gamma_{0i}^k}{\sum_{k=1}^s \sum_{i=1}^m \gamma_{0i}^k}, k = 1, \dots, s. \quad (20)$$

C. Determination of Attribute Weights Based on Regret Theory

Based on decision makers' regret psychological behavior, the perceived utility of attribute for decision makers is

$$u_{ij}^k = v(\hat{r}_{ij}^k) + R[v(\hat{r}_{ij}^k) - v(\hat{r}_{ij}^{k*})] \quad (21)$$

$i = 1, \dots, m, j = 1, \dots, n, k = 1, \dots, s$

where $\hat{r}_{ij}^{k*} = \max\{\hat{r}_{ij}^k | i = 1, 2, \dots, m\}$, power function is used as utility function of attribute value in this article, γ is the risk avoidance coefficient of decision maker, and the smaller γ is, the greater the degree of risk avoidance of decision maker [57], that is

$$v(\hat{r}_{ij}^k) = (\hat{r}_{ij}^k)^\gamma, \gamma \in (0, 1), i = 1, \dots, m, j = 1, \dots, n, k = 1, \dots, s. \quad (22)$$

The value of parameter γ adopts the data obtained by experimental verification: $\gamma = 0.88$ [57]. On this basis, the regret value of each scheme for each attribute relative to the ideal point in each state is calculated. Therefore, the fuzzy intuitive perception utility function can be expressed as

$$u_{ij}^k = (\hat{r}_{ij}^k)^\gamma + 1 - \exp\{-\delta((\hat{r}_{ij}^k)^\gamma - (\hat{r}_{i*}^k)^\gamma)\} \quad (23)$$

$i = 1, \dots, m, j = 1, \dots, n, k = 1, \dots, s.$

δ is called the regret avoidance coefficient of decision makers, the greater the degree of regret avoidance of decision makers, the larger the δ ($\delta > 0$). And the value of parameter $\delta = 0.3$ [22] is obtained by experimental verification.

Where

$$\hat{r}_{i*}^k = \max_{j=1,2,\dots,n} \{\hat{r}_{ij}^k\}, j = 1, \dots, n, k = 1, \dots, s. \quad (24)$$

Thus, the utility matrix $U^{(k)}$ can be established as

$$U^{(k)} = (u_{ij}^k)_{m \times n}, k = 1, \dots, s. \quad (25)$$

In order to ensure the maximum total regret perceived utility of decision makers, based on the construction of the total regret utility function, it is necessary to ensure the optimal allocation of attribute weights, the optimal model is constructed as follows:

$$\begin{aligned} \max \sum_{i=1}^m u_i &= \sum_{i=1}^m \sum_{k=1}^s \lambda_k \sum_{j=1}^n \omega_j u_{ij}^k \\ \text{s.t. } \begin{cases} u_{ij}^k = (s(r_{ij}^k))^\gamma + 1 - \exp\{-\delta((s(r_{ij}^k))^\gamma - (s(r_{i*j}^k))^\gamma)\} \\ \sum_{j=1}^n \omega_j = 1, \omega_j \in (0, 1) \\ i = 1, \dots, m, j = 1, \dots, n, k = 1, \dots, s. \end{cases} \end{aligned} \quad (26)$$

The objective function aims to maximize the total utility u_i , which represents the aggregated perceived utility of the decision-makers. The model's first constraint establishes the methodology for calculating perceived utility. The second constraint guarantees that the total of all attribute weights equals 1, ensuring that each weight falls within the (0,1) interval. The third constraint specifies the parameter count.

Theorem 4: The model (26) has an optimal solution.

Proof: The objective function is composed of continuous functions, so the continuity of the whole objective function is guaranteed. Second, considering that the variables w_j , λ_k , and u_{ij}^k involved in the objective function vary in a limited range, the boundedness of the objective function is ensured.

The constraint conditions are confined to a closed and bounded set, which not only ensures that the bounded space is understood, but also ensures that it is closed. According to Weierstrass approximation theorem [58], a continuous function defined on a closed and bounded set must take the maximum and minimum values on that set. In view of the continuous and bounded properties of the objective function and the closed solution space defined by constraints, it is evident that the model (26) necessarily has an optimal solution. $\square$

D. Interference Calculation of MAFQGDM Problem Based on Deng Entropy

The MAFQGDM model proposes a practical approach for meteorological disaster emergency rescue by merging quantum decision theory with a Bayesian decision-making framework. This model offers an innovative tool for planning and implementing emergency responses, utilizing a three-layer Bayesian network to capture and describe the intricate relationships among decision-makers, meteorological elements, and rescue alternatives. The initial layer represents decision-making groups, consisting of rescue experts whose evaluations are influenced by their individual assessments and interactions. The intermediate layer includes various meteorological factors crucial to the decision-making process, impacting the feasibility and effectiveness of rescue alternatives. This layer ensures all relevant environmental conditions are systematically incorporated into the decision-making process. The final layer comprises feasible

rescue plans, influenced by both the evaluations from decision-making groups and the state of meteorological elements. By integrating quantum decision theory, the model addresses the probabilistic and uncertain nature of decision-making in emergency scenarios. This dynamic and comprehensive representation ensures a robust and responsive framework for emergency decision-making, capturing mutual interference effects between decision-makers and providing a detailed analysis of the influence of meteorological factors.

Decision-makers' weights constitute the first layer probabilities of BN

$$P(d_k) = \lambda_k, k = 1, \dots, s. \quad (27)$$

The regret utility values and attributes' weights are used to get the conditional probabilities in BN

$$P(a_i|d_k) = \frac{e^{\sum_{j=1}^n \omega_j u_{ij}^k}}{\sum_{i=1}^m e^{\sum_{j=1}^n \omega_j u_{ij}^k}}, i = 1, \dots, m, k = 1, \dots, s. \quad (28)$$

Traditional phase angle difference estimation methods are subjective and prone to errors. Entropy is highly effective in handling uncertain information, but traditional Shannon entropy has limitations when dealing with incomplete information and fuzzy decision-making. In contrast, Deng entropy offers significant advantages in measuring uncertainty, particularly in decision-making situations involving incomplete information. Deng entropy not only reflects the amount of information but also adapts more effectively to fuzzy and uncertain decision environments [48]. Furthermore, Deng entropy has been widely applied in multiattribute decision-making, particularly in measuring interference effects, demonstrating its effectiveness and applicability [36], [38]. Therefore, this article proposes using Deng entropy to measure the interference angle.

First, the decision maker's preference matrix for alternatives $T = (t_{ik})_{m \times s}$ is obtained based on the BN. t_{ik} ($i = 1, 2, \dots, m, k = 1, 2, \dots, s$) represents the decision maker's preferences for the alternatives a_i ($i = 1, 2, \dots, m$) and satisfy $\sum_{i=1}^m \sum_{k=1}^s t_{ik} = 1$, calculated as follows:

$$t_{ik} = P(d_k)P(a_i|d_k), i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (29)$$

On this basis, the quantum preference matrix $T' = (t'_{ik} e^{i\theta_{ik}})_{m \times s}$ is computed, and the probability is determined by squaring the length of the projectio.

$$t'_{ik} = \sqrt{t_{ik}}, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (30)$$

Second, a connection matrix consisting of connection vectors [38] $\vec{\gamma}_{i,e}$ ($i = 1, 2, \dots, m, e = 1, 2, \dots, \frac{s(s-1)}{2}$) is constructed by combining pairs of the same choices under different decision-makers

$$\begin{aligned} \vec{\gamma}_{i,1} &= \begin{bmatrix} \vartheta_{i1} \\ \vartheta_{i1} \end{bmatrix} = \begin{bmatrix} \vartheta_{i1} \\ \vartheta_{i2} \end{bmatrix}, \vec{\gamma}_{i,2} = \begin{bmatrix} \vartheta_{i2} \\ \vartheta_{i2} \end{bmatrix} = \begin{bmatrix} \vartheta_{i1} \\ \vartheta_{i3} \end{bmatrix}, \dots, \\ \vec{\gamma}_{i,n-1} &= \begin{bmatrix} \vartheta_{is} \\ \vartheta_{is} \end{bmatrix} = \begin{bmatrix} \vartheta_{i1} \\ \vartheta_{is} \end{bmatrix} \\ &\dots \end{aligned}$$

$$\vec{\gamma}_{i \frac{s(s-1)}{2}} = \begin{bmatrix} v_{i \frac{s(s-1)}{2}} \\ \vartheta_{i \frac{s(s-1)}{2}} \end{bmatrix} = \begin{bmatrix} t_{is-1} \\ t_{is} \end{bmatrix}, i = 1, 2, \dots, m \quad (31)$$

where $e(1, 2, \dots, \frac{s(s-1)}{2})$ is the number reassigned to the association between decision-makers. For example, when $e = 1$, $\vec{\gamma}_{i,1}$ denotes the connection vector between the first decision maker and the second decision maker.

Then, the confidence distance is calculated

$$M_{ik\hat{k}} = \left| v_{ik} + \frac{|v_{ik} - \eta| - |\vartheta_{ik} - \eta|}{|v_{ik} - \eta| + |\vartheta_{ik} - \eta|} \right|$$

$$k = 1, \dots, s-1, \hat{k} = k+1, \dots, s, i = 1, 2, \dots, m. \quad (32)$$

If $|v_{ik} - \eta| \leq |\vartheta_{ik} - \eta|$, then the positions of $|v_{ik} - \eta|$ and $|\vartheta_{ik} - \eta|$ need to be reversed, and $\eta = \frac{1}{ms}$.

On this basis, the Deng entropy between vectors $\vec{v}_k(k = 1, \dots, s-1)$ and $\vec{\vartheta}_{\hat{k}}(\hat{k} = k+1, \dots, s)$ is calculated

$$D_{ik\hat{k}} = M_{ik\hat{k}} \log_2 \frac{M_{ik\hat{k}}}{2L-1}$$

$$k = 1, \dots, s-1, \hat{k} = k+1, \dots, s, i = 1, 2, \dots, m. \quad (33)$$

In order to ensure that the value range of the Deng entropy measure conforms to the value range of $\cos(\theta_{d_k}^i - \theta_{d_{\hat{k}}}^i)$ in $[-1, 1]$, where $\bar{D}'_{ik\hat{k}}$ represents the average of the columns in which the confidence distance $D'_{ik\hat{k}}$ is located, the Deng entropy is normalized

$$E_{ik\hat{k}} = \frac{D'_{ik\hat{k}} - \bar{D}'_{ik\hat{k}}}{\max_i \{D'_{ik\hat{k}}\} - \min_i \{D'_{ik\hat{k}}\}}$$

$$k = 1, \dots, s-1, \hat{k} = k+1, \dots, s, i = 1, 2, \dots, m. \quad (34)$$

The cosine of the phase difference is defined by $E_{ik\hat{k}}$

$$\cos(\theta_{d_k}^i - \theta_{d_{\hat{k}}}^i) = E_{ik\hat{k}}$$

$$k = 1, \dots, s-1, \hat{k} = k+1, \dots, s, i = 1, 2, \dots, m. \quad (35)$$

Then, the interference effect can be expressed as

Interference_{*i*}

$$= 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s \varphi_{d_k} \varphi_{a_i|d_k} \varphi_{d_{\hat{k}}} \varphi_{a_i|d_{\hat{k}}} \cos(\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i)$$

$$= 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s \varphi_{d_k} \varphi_{a_i|d_k} \varphi_{d_{\hat{k}}} \varphi_{a_i|d_{\hat{k}}} E_{ik\hat{k}}$$

$$i = 1, 2, \dots, m. \quad (36)$$

The cosine of the interference angle ($\cos(\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i)$) provides insights into the nature of the interference.

1) Interference angle $\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i = 0^\circ$: When the interference angle is 0° , $\cos(\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i) = 1$, indicating a

strong positive interference. This means that the decision-makers' evaluations are highly aligned, reinforcing each other's preferences.

- 2) Interference angle $\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i = 90^\circ$: At an angle of 90° , $\cos(\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i) = 0$, suggesting that the decision-makers' evaluations are independent of each other. There is no interference effect, and their preferences do not influence one another.
- 3) Interference angle $\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i = 180^\circ$: When the interference angle is 180° , $\cos(\theta_{a_i|d_k}^i - \theta_{a_i|d_{\hat{k}}}^i) = -1$, indicating a strong negative interference. This means that the decision-makers' evaluations are in complete opposition, directly counteracting each other's preferences.

Similarly, the closer $|E_{ik\hat{k}}|$ approaches 1, the greater the interference intensity, and the positive or negative of $E_{ik\hat{k}}$ represents the direction of interference.

Finally, the quantum probability value of each alternative is calculated

$$P(a_i) = \sigma \left(\sum_{k=1}^s \left| \varphi_{d_k} e^{i\theta_{d_k}} \varphi_{a_i|d_k} e^{i\theta_{a_i|d_k}} \right|^2 + \text{interference}_i \right)$$

$$i = 1, 2, \dots, m \quad (37)$$

where

$$\sigma = \frac{1}{\sum_{i=1}^m \left| \sum_{k=1}^s \varphi_{d_k} \varphi_{a_i|d_k} \right|^2}. \quad (38)$$

Then, the quantum probability of alternative is calculated and ranked.

E. Decision Steps

The decision steps of the MAFQGDM model considering interference effects proposed in this article are shown in Fig. 5, and the detailed contents are as follows.

Step 1: Determine the evaluation information. Each decision maker $d_k(k = 1, \dots, s)$ evaluates attribute $c_j(j = 1, \dots, n)$ about alternative $a_i(i = 1, \dots, m)$ and obtains the evaluation information matrix $R^{(k)} = (r_{ij}^k)_{m \times n}$ using intuitionistic fuzzy numbers.

Step 2: Normalize the score function matrix. According to (9), the initial information evaluation matrix is transformed into score function matrix $\hat{R}^{(k)} = (s(r_{ij}^k))_{m \times n}$, and get the normalized matrix $\hat{R}^{(k)} = (\hat{r}_{ij}^k)_{m \times n}$ according to (13).

Step 3: Calculate the weights of decision makers by using the grey correlation degree. The decision maker's weight λ_k is determined using the (14)–(20).

Step 4: Determine attributes' weights by combining regret theory. Based on the score function, the fuzzy intuitive perception utility value u_{ij}^k is calculated according to (23), and the optimal attribute weight vector is obtained by the model (26).

Step 5: Determine the QLBN. The decision maker's weight $\lambda_k(k = 1, \dots, s)$ is determined as the node probability $P(d_k), k = 1, \dots, s$. The conditional probability $P(a_i|d_k), i = 1, \dots, m, k = 1, \dots, s$ in the Bayesian network is obtained by (27).

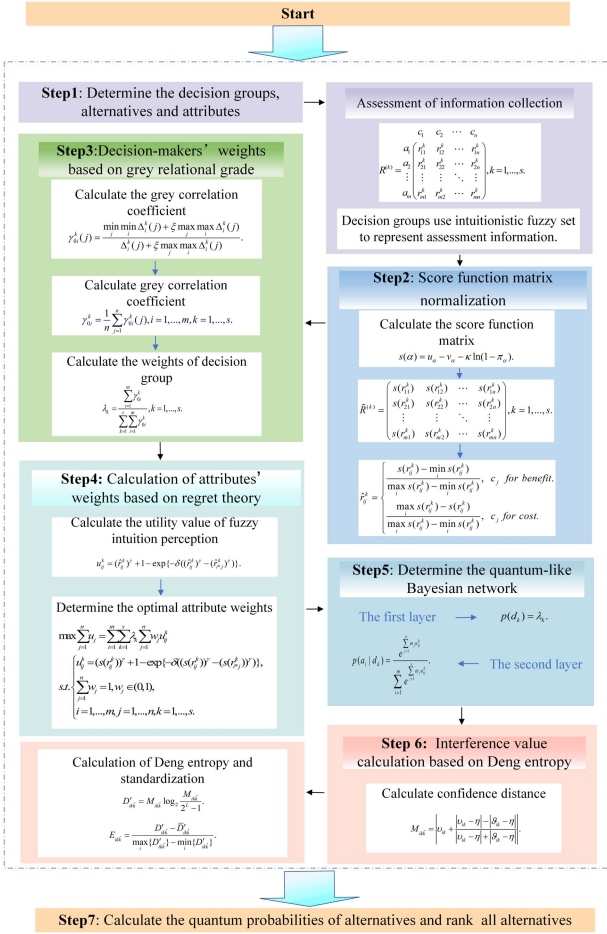


Fig. 5. MAFQGDM model decision steps.

Step 6: Interference value calculation based on Deng entropy. The confidence distance is calculated by (32), the entropy is calculated and normalized according to (33) and (34), and the cosine of the phase difference is calculated using (35).

Step 7: The quantum probability $P(a_i), i = 1, \dots, m$ of rescue alternative is obtained by (37), and the alternatives are ranked by the quantum probability values.

IV. CASE ANALYSIS

In this section, a meteorological disasters rescue is used to verify the feasibility and effectiveness of the proposed MAFQGDM.

A. Background Introduction

During the emergency response to meteorological disasters, it is crucial to develop and choose efficient rescue strategies. This article focusses on the region of Zhuozhou and its neighboring regions in Hebei Province, China, during the month of July in 2023. Abnormal rainfall is responsible for a range of calamities, including urban waterlogging, flash floods, mudslides, and building collapses. These calamities pose a direct threat to the stability of the local community and the safety of the general public. Creating response plan necessitates a precise evaluation

of the magnitude and consequences of the catastrophe to enable efficient allocation of resources and evacuation of victims, hence facilitating appropriate medical care and evacuation. Based on this, this article puts forward five different emergency rescue plans, involving flood rescue measures, emergency transfer of residents, distribution of relief materials, strategic deployment of rescue teams and other key links, covering the following five main attributes.

- 1) *Casualty level* c_1 : This attribute refers to the number and severity of injuries and deaths in situations such as natural disasters, accidents, conflicts, etc.
- 2) *Degree of economic loss* c_2 : This attribute measures the damage caused by an event to economic indicators such as material property, productive capacity, and market stability.
- 3) *Precision facility resilience* c_3 : This attribute is mainly used to assess the resilience of critical infrastructure and important services after disasters. These sophisticated facilities may include medical centers, emergency services, etc.
- 4) *Traffic disruption scope* c_4 : This attribute focusses on the influence of the event on the transportation infrastructure, including the disruption of transportation lines, damage to vehicles, and reduced transportation capacity.
- 5) *Communication outage time* c_5 : This attribute refers to the speed and efficiency with which rescue workers, resources and services can reach the affected area during a disaster or emergency.

There are three decision groups $d_k (k = 1, 2, 3)$ to evaluate emergency rescue alternatives $a_i (i = 1, \dots, 5)$. In the process of emergency rescue, it is often associated with disaster. At the same time, the decision maker's evaluation and selection of the scheme will also be disturbed and affected by other decision makers, so this article uses the quantum group decision model to select the emergency rescue alternatives.

B. Decision Steps

Step 1: Collect the assessment information. The evaluation information of the three decision makers is as follows:

$$R^{(1)} = \begin{pmatrix} (0.5, 0.5) & (0.4, 0.2) & (0.5, 0.4) & (0.4, 0.5) & (0.6, 0.1) \\ (0.1, 0.7) & (0.4, 0.3) & (0.2, 0.4) & (0.6, 0.3) & (0.7, 0.2) \\ (0.4, 0.2) & (0.3, 0.1) & (0.8, 0.1) & (0.6, 0.2) & (0.1, 0.1) \\ (0.5, 0.4) & (0.5, 0.5) & (0.6, 0.3) & (0.2, 0.7) & (0.3, 0.5) \\ (0.2, 0.5) & (0.4, 0.3) & (0.5, 0.2) & (0.1, 0.7) & (0.6, 0.3) \end{pmatrix}$$

$$R^{(2)} = \begin{pmatrix} (0.6, 0.3) & (0.5, 0.2) & (0.6, 0.1) & (0.8, 0.1) & (0.7, 0.3) \\ (0.4, 0.6) & (0.5, 0.5) & (0.6, 0.2) & (0.2, 0.6) & (0.4, 0.4) \\ (0.5, 0.5) & (0.4, 0.6) & (0.6, 0.3) & (0.5, 0.2) & (0.6, 0.3) \\ (0.2, 0.1) & (0.4, 0.3) & (0.4, 0.5) & (0.2, 0.7) & (0.3, 0.3) \\ (0.6, 0.1) & (0.8, 0.2) & (0.4, 0.1) & (0.5, 0.3) & (0.6, 0.1) \end{pmatrix}$$

$$R^{(3)} = \begin{pmatrix} (0.9, 0.1) & (0.7, 0.1) & (0.6, 0.3) & (0.5, 0.1) & (0.8, 0.2) \\ (0.6, 0.4) & (0.4, 0.5) & (0.3, 0.6) & (0.1, 0.4) & (0.5, 0.5) \\ (0.6, 0.2) & (0.5, 0.3) & (0.7, 0.2) & (0.4, 0.6) & (0.3, 0.1) \\ (0.1, 0.1) & (0.4, 0.5) & (0.3, 0.1) & (0.7, 0.2) & (0.6, 0.3) \\ (0.5, 0.4) & (0.6, 0.2) & (0.4, 0.4) & (0.4, 0.5) & (0.6, 0.1) \end{pmatrix}.$$

Step 2: Get the score function matrix and normalize it. First, the initial information evaluation matrix is transformed into the score function matrix $\tilde{R}^{(k)} = (s(r_{ij}^k))_{m \times n}$, then each score function matrix $\tilde{R}^{(k)} = (s(r_{ij}^k))_{m \times n}$ is normalized, and finally the normalized matrix $\hat{R}^{(k)} = (\hat{r}_{ij}^k)_{m \times n}$ is obtained

$$\hat{R}^{(1)} = \begin{pmatrix} 0.0703 & 0.0000 & 0.4121 & 0.4565 & 0.1119 \\ 1.0000 & 0.9431 & 0.0000 & 0.0565 & 0.0000 \\ 0.0276 & 1.0000 & 1.0000 & 0.0000 & 1.0000 \\ 0.0000 & 0.4760 & 0.6080 & 0.8565 & 0.9964 \\ 0.6503 & 0.9431 & 0.5262 & 1.0000 & 0.2680 \end{pmatrix}$$

$$\hat{R}^{(2)} = \begin{pmatrix} 0.2048 & 0.5390 & 1.0000 & 0.0000 & 0.0000 \\ 1.0000 & 0.7500 & 0.8838 & 0.9529 & 0.8645 \\ 0.6484 & 1.0000 & 0.7744 & 0.4029 & 0.2599 \\ 0.9390 & 0.7890 & 0.0000 & 1.0000 & 1.0000 \\ 0.0000 & 0.0000 & 0.4743 & 0.4529 & 0.0549 \end{pmatrix}$$

$$\hat{R}^{(3)} = \begin{pmatrix} 0.0000 & 0.0000 & 0.7500 & 0.2309 & 0.0000 \\ 0.5485 & 1.0000 & 0.0000 & 1.0000 & 0.9449 \\ 0.4490 & 0.6093 & 1.0000 & 0.6830 & 1.0000 \\ 1.0000 & 1.0000 & 0.3909 & 0.0000 & 0.5476 \\ 0.6835 & 0.3046 & 0.3206 & 0.6283 & 0.3641 \end{pmatrix}.$$

Step 3: Determine decision maker's weights based on grey correlation degree. Let the resolution coefficient be $\xi = 0.5$, the weight of each decision maker can be calculated by the (19)–(25), and the vector is

$$\Lambda = \{0.3555, 0.3365, 0.3080\}.$$

Step 4: Calculate attribute weights based on regret theory. The perceived utility matrix $U^{(k)} (k = 1, 2, 3)$ can be obtained according to (23)

$$U^{(1)} = \begin{pmatrix} -0.2146 & -0.3499 & 0.2819 & 0.3402 & -0.1467 \\ 1 & 0.9346 & -0.3499 & -0.2382 & -0.3499 \\ -0.2903 & 1 & 1 & -0.3499 & 1 \\ -0.3499 & 0.3656 & 0.5332 & 0.8336 & 0.9959 \\ 0.5856 & 0.9346 & 0.4301 & 1 & 0.0853 \end{pmatrix}$$

$$U^{(2)} = \begin{pmatrix} -0.0055 & 0.4464 & 1 & -0.3499 & -0.3499 \\ 1 & 0.7069 & 0.8656 & 0.9459 & 0.8430 \\ 0.5832 & 1 & 0.7362 & 0.2697 & 0.0739 \\ 0.9298 & 0.7537 & -0.3499 & 1 & 1 \\ -0.3499 & -0.3499 & 0.3634 & 0.3355 & -0.2410 \end{pmatrix}$$

$$U^{(3)} = \begin{pmatrix} -0.3499 & -0.3499 & 0.7069 & 0.3525 & -0.3499 \\ 0.4584 & 1 & -0.3499 & 1 & 0.9366 \\ 0.3305 & 0.5348 & 1 & 0.6257 & 1 \\ 1 & 1 & 0.2537 & -0.3499 & 0.4573 \\ 0.6263 & 0.1365 & 0.1585 & 0.5584 & 0.2178 \end{pmatrix}.$$

The attributes' weighting vector is calculated by model (26)

$$W = \{0.2259, 0.1983, 0.1493, 0.1840, 0.2425\}.$$

Step 5: Determine the QLBN. The weights of the decision groups are determined as the node probability in the first layer of the QLBN

$$P(d_1) = 0.3555, \quad P(d_2) = 0.3365, \quad P(d_3) = 0.3080.$$

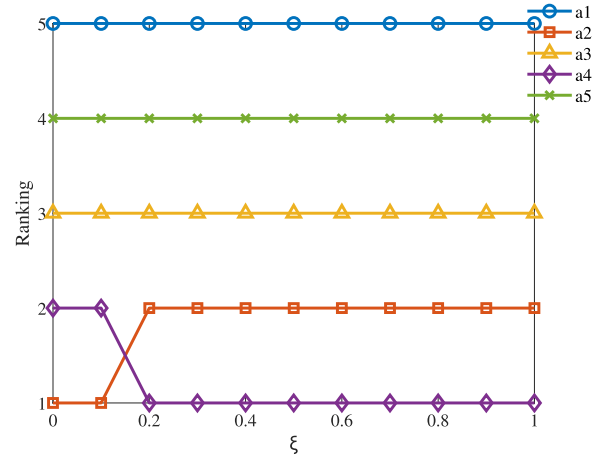


Fig. 6. Sensitivity analysis based on ξ .

The conditional probability in QLBN is obtained by (28)

$$\begin{aligned} P(a_1|d_1) &= 0.1325, & P(a_2|d_1) &= 0.1751, & P(a_3|d_1) &= 0.2203 \\ P(a_4|d_1) &= 0.2221, & P(a_5|d_1) &= 0.2500, & P(a_1|d_2) &= 0.1340 \\ P(a_2|d_2) &= 0.2942, & P(a_3|d_2) &= 0.2040, & P(a_4|d_2) &= 0.2557 \\ P(a_5|d_2) &= 0.1121, & P(a_1|d_3) &= 0.1122, & P(a_2|d_3) &= 0.2454 \\ P(a_3|d_3) &= 0.2521, & P(a_4|d_3) &= 0.2108, & P(a_5|d_3) &= 0.1795. \end{aligned}$$

Step 6: Calculate interference values based on Deng entropy. The standardized Deng entropy is calculated according to (33) and (34)

$$E_{12} = \begin{bmatrix} -0.5382 \\ 0.4618 \\ 0.4420 \\ -0.1100 \\ -0.2555 \end{bmatrix}, \quad E_{13} = \begin{bmatrix} -0.1181 \\ 0.0485 \\ -0.4994 \\ 0.5006 \\ 0.0685 \end{bmatrix}, \quad E_{23} = \begin{bmatrix} -0.6134 \\ 0.1298 \\ 0.2454 \\ 0.3866 \\ -0.1485 \end{bmatrix}.$$

Step 7: The quantum probability of each alternative is obtained using (37)

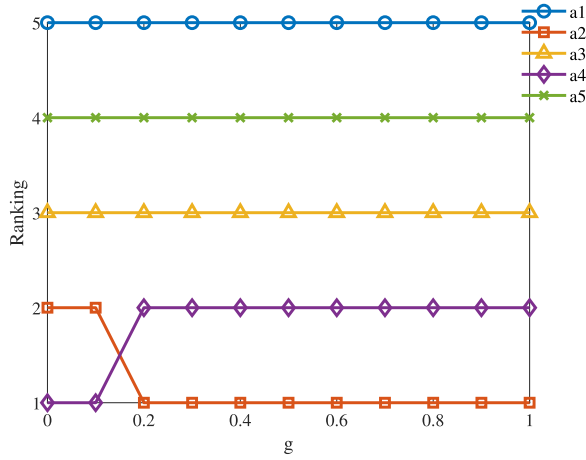
$$\begin{aligned} P(a_1) &= 0.0175, & P(a_2) &= 0.3092, & P(a_3) &= 0.2260. \\ P(a_4) &= 0.3118, & P(a_5) &= 0.1355. \end{aligned}$$

The alternatives are sorted according to quantum probability values, $a_4 \succ a_2 \succ a_3 \succ a_5 \succ a_1$.

C. Sensitivity Analysis

1) This article studies the effect of resolution coefficient related to grey correlation degree on rescue plan ranking. By adjusting the resolution coefficient, it is possible to understand not only the change in the weight of decision makers, but also the effectiveness of the proposed model. Based on this, a sensitivity analysis was performed to determine the precise effect of resolution factor adjustments on the ranking of alternatives. The results are shown in Fig. 6.

With the change in ξ , the ranking order of the alternatives a_2 and a_4 has changed, while the ranking of the other alternatives remains the same. The resolution coefficient ξ mainly affects the calculation of grey correlation degree, and then affects the

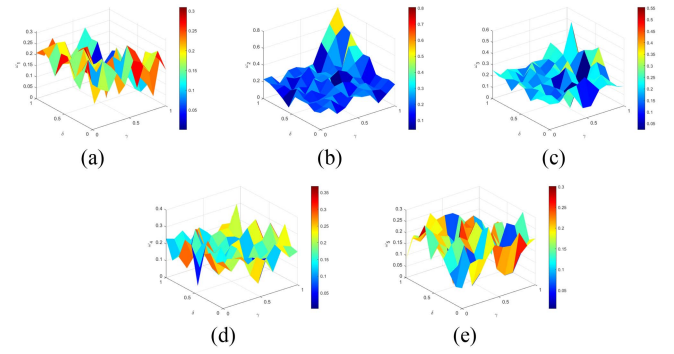

 Fig. 7. Sensitivity analysis based on g .

weights of decision makers. The quantum framework reduces the influence of single decision maker weight on the overall decision result by introducing the interference effect between decision makers, that is, the introduction of phase angle. This shows that the quantum framework has high robustness and adaptability in the decision environment with strong uncertainty, complexity and fuzziness.

2) Consider the influence of decision maker's subjective and objective weight coefficient g on the ranking of alternatives. Assuming that the top decision maker assigns a subjective weight $\lambda^0 = (0.55, 0.30, 0.15)$ to the three decision makers, the comprehensive weight of decision makers can be calculated according to the calculation formula: $\hat{\lambda} = g\lambda^0 + (1 - g)\lambda$, considering $g \in [0, 1]$, and the ranking change results of alternative schemes are shown in Fig. 7.

The analysis shown in Fig. 7 shows that as the weight coefficient g changes, only the ranking positions of alternatives a_2 and a_4 fluctuate, while the ranking of the other alternatives remains stable. This result further validates the robustness of the model, especially in a decision environment with high uncertainty, where the fluctuation of decision maker's weights does not have a significant impact on the ranking of all alternatives. The robustness of the quantum decision framework stems from its ability to effectively deal with interference effects among decision makers. When the weight coefficient g changes, the quantum model quantizes the interference effect between different decision makers through the phase angle, which makes the weight change of a single decision maker not significantly affect the final decision result.

3) *Sensitivity analysis of parameters δ and γ in regret theory:* In the actual decision-making process, decision makers are usually faced with complex and changeable decision-making environment, and different decision-making scenarios may involve different risk preferences, information uncertainty, and situation changes, which makes the parameters of the model have greater flexibility. Therefore, in order to make the model adapt to different decision needs and accurately reflect the behavior characteristics of decision makers, In the actual decision-making process, decision makers are usually faced with complex and changeable


 Fig. 8. Influence of parameters δ and γ on the weight of each attribute. (a) Attribute's weight ω_1 . (b) Attribute's weight ω_2 . (c) Attribute's weight ω_3 . (d) Attribute's weight ω_4 . (e) Attribute's weight ω_5 .

decision-making environment, and different decision-making scenarios may involve different risk preferences, information uncertainty and situation changes, which makes the parameters of the model have greater flexibility. Therefore, in order to make the model adapt to different decision requirements and accurately reflect the behavior characteristics of decision makers, this article considers the value of δ as 0-1 and γ as 0-1, and observes the influence of different parameter combinations on the weight of each attribute, as shown in Fig. 8.

It can be seen from the Fig. 8 that with the increase of δ , the fluctuation of attribute weight significantly increases, especially in attribute c_2 , which shows an obvious nonlinear growth trend, indicating that regret effect plays a more significant role in this attribute. When γ is high, the model is more inclined to strengthen the weight of efficient use value attributes and weaken the influence of inefficient use value attributes, which shows complex nonlinear fluctuations in the weight changes of other attributes. Combined with the background of emergency decision of meteorological disaster, the model can flexibly cope with the complex and changeable attribute priority adjustment requirements by adjusting the values of δ and γ . For example, in resource allocation decisions, the importance of different attributes may change dynamically over time, and appropriate parameter adjustment can more accurately reflect this change while ensuring the robustness of the model to different scenarios. Comprehensive analysis of the image results, choosing the parameters in the medium range (such as $\delta = 0.3, \gamma = 0.88$) can achieve a better balance between regret effect and utility sensitivity. This parameter adjustment mechanism further verifies the applicability and flexibility of the model in high uncertainty decision environment.

D. Comparative Analysis

In order to verify the effectiveness of the proposed method, a comparison is made with several existing decision models. These comparison models can be implemented directly or adjusted indirectly to address the case issues. The ranking values of each plan obtained by each model are shown in Table I. Compared with the traditional models, the ranking obtained by the proposed method is significantly different. For example, The optimal alternative of Classic-TOPSIS [59] method is a_4 , but the

TABLE I
COMPARISON OF RANKING RESULTS OF DIFFERENT MODELS

Methods	Ranking indices	Final ranking orders
Classic -TOPSIS [59]	$C_1^* = 0.1788, C_2^* = 0.6740,$ $C_3^* = 0.6790, C_4^* = 0.7086,$ $C_5^* = 0.3555.$	$a_4 \succ a_3 \succ a_2$ $\succ a_5 \succ a_1$
An extended TODIM [60]	$\phi(x_1) = -2.2441, \phi(x_2) = 1.0979,$ $\phi(x_3) = 0.7462, \phi(x_4) = 0.7098,$ $\phi(x_5) = -0.3098.$	$a_2 \succ a_3 \succ a_4$ $\succ a_5 \succ a_1$
Quantum model 1 [30]	$P(a_1) = 0.0021, P(a_2) = 0.3764,$ $P(a_3) = 0.2015, P(a_4) = 0.2522,$ $P(a_5) = 0.1678.$	$a_2 \succ a_4 \succ a_3$ $\succ a_5 \succ a_1$
Quantum model 2 [31]	$P(a_1) = 0.2189, P(a_2) = 0.0798,$ $P(a_3) = 0.3713, P(a_4) = 0.3948,$ $P(a_5) = 0.0661.$	$a_4 \succ a_3 \succ a_1$ $\succ a_2 \succ a_5$
The proposed method in BN	$P(a_1) = 0.1267, P(a_2) = 0.2368,$ $P(a_3) = 0.2246, P(a_4) = 0.2299,$ $P(a_5) = 0.1819.$	$a_2 \succ a_4 \succ a_3$ $\succ a_5 \succ a_1$
The proposed method	$P(a_1) = 0.0175, P(a_2) = 0.3092,$ $P(a_3) = 0.2260, P(a_4) = 0.3118,$ $P(a_5) = 0.1355.$	$a_4 \succ a_2 \succ a_3$ $\succ a_5 \succ a_1$

sorting results of other schemes are significantly different from those in this article. TOPSIS method is mainly calculated based on the relative proximity of the scheme, and does not consider the mutual interference effect between decision makers, so its adaptability in complex decision-making scenarios is limited. The optimal scheme of TODIM method [60] is a_2 , and its theoretical basis is cumulative prospect theory. However, this method fails to dynamically adjust decision weights, nor does it reflect the influence of decision maker's psychological state on the ranking results. In the model proposed in this article, the optimal scheme obtained under the BN is a_2 , and when extended to the QLBN, the optimal alternative becomes a_4 . This difference is mainly due to the fact that the traditional model ignores the interference effect between decision makers, but this article effectively quantifies the interaction between decision makers by introducing a quantum decision framework, thus dynamically adjusting the ordering results.

In comparison with other quantum models, the optimal choice of quantum model 1 [30] is a_2 , but its attributes' weights and decision makers' weights are subjective and lack dynamic adaptability, especially in uncertain and complex decision environments. In contrast, the model in this article determines the weights of decision makers dynamically through the grey relational degree method, and constructs an optimization model to solve the attributes' weights by combining the regret theory, taking into account the influence of the psychological state of decision makers, making the weight distribution more reasonable and adaptable to the complex decision-making environment. The optimal alternative of quantum model 2 [31] is a_4 , but the ordering of other alternatives is significantly different from the model presented in this article. The main reason is that the model has some errors in calculating phase angle, which affects the accurate description of quantum interference effect. In this article, the Deng Entropy-based method is used to accurately measure the interference effect among decision makers, and the ranking results are dynamically adjusted in combination with regret psychology. At the same time, the IFS is used as the

information representation method to describe the ambiguity and uncertainty in the complex decision environment more comprehensively. This design enhances the robustness and adaptability of the model, which makes the model more efficient and reliable when dealing with complex fuzzy information.

In conclusion, compared with traditional models and other quantum models, the proposed model has more obvious advantages in dealing with uncertainty, fuzziness, and interference effects of decision makers. By constructing a quantum-like Bayesian framework, combining Deng entropy, regret theory, and IFS, the model not only fits the complex decision environment better in terms of ordering results, but also embodies stronger theoretical rigor and application flexibility.

V. CONCLUSION

In order to address the complexity, dynamics, and uncertainty of meteorological disasters, a fuzzy quantum group decision-making model is proposed by combining quantum decision theory, regret theory, grey relational degree, and Deng entropy, thereby leveraging the synergistic effect of their integration. First, IFS are used to represent the evaluation information. Second, the correlation between decision makers and group consensus is quantified using the grey relational degree method to allocate decision makers' weights appropriately. In addition, regret theory is introduced to dynamically adjust attributes' weights, fully reflecting the psychological and behavioral characteristics of decision makers facing potential regret, and enhancing the model's adaptability to dynamic changes in decision making. Finally, a QLBN is constructed to address the interference effect among decision makers, and Deng entropy is used to measure specific interference values, accurately depicting the complex interactions among decision makers. Comparative experiments show that the integration of these methods significantly improves the model's adaptability, stability, and efficiency in selecting complex emergency alternatives. The decision-making results are more accurate and robust than those obtained using any single method. Compared to existing quantum decision models, MAFQGDM considers both attributes' weights and decision makers' weights. MAFQGDM integrates grey correlation analysis and regret theory into a quantum-like Bayesian framework of fuzzy uncertainty, making it a more comprehensive multi-attribute quantum group decision-making method. Therefore, the model presented in this article has strong generality to multiattribute group decision making problems with uncertainty and fuzziness. Uncertainty and fuzziness are common features of decision-making, whether in meteorological disaster emergency decision-making scenarios or in other areas such as medical resource allocation, supply chain optimization or policy making.

In summary, the quantum group decision model offers a comprehensive framework that integrates psychological factors and mutual interference among decision-makers. This approach enhances the realism and reliability of decision outcomes and provides valuable insights into the robustness of the model in real-world environments, particularly under high-risk conditions such as meteorological disasters. To ensure computational efficiency and a detailed analysis of interference effects, we have

decided to limit the number of decision makers to 8 or fewer, consistent with the statistical definition of a small sample size. This decision is informed by the formula for potential interference effects $\frac{s(s-1)}{2}$, which helps us prevent an unsustainable computational burden. In contrast, the increase of the number of alternatives m and the number of attributes n has a limited effect on the computational complexity of the model. Increasing the number of alternatives and attributes will increase the dimensions of the decision maker's evaluation information matrix, but this expansion will only lead to a linear increase in computation. In addition, future research directions can consider the following two aspects: 1) Quantum-like Bayesian framework has advantages in dealing with decision-maker interference effects, but its application in high-dimensional multiattribute decision problems may face the challenge of increasing computational complexity. As the number of decision-makers and attributes increases, the quantum state dimension will grow exponentially, potentially leading to reduced computational efficiency. 2) The real-time adaptability of the model in the dynamic decision-making environment. In dynamic decision environment, frequent data update may cause the model to need to adjust the weight and recalculate the utility function several times, which increases the difficulty of real-time calculation. In addition, the superposition of incomplete information and complex interference effects will aggravate the uncertainty of decision-making.

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